

Effectiveness of intraarterial administration of fasudil hydrochloride for preventing symptomatic vasospasm after subarachnoid hemorrhage.

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We examined the effect of intraarterial administration of fasudil hydrochloride (IAFC), a Rho kinase inhibitor, for the prevention of symptomatic vasospasm after SAH by evaluating cerebral circulation. We evaluated IAFC cases of 57 sides of 38 patients (12 men and 26 women, average age 60.2 years old) diagnosed with aneurysmal subarachnoid hemorrhage (SAH) from February 2012 to November 2012. All cases were treated by clipping or coil embolization within 48 h after onset. Indication for IAFC was the existence of a spastic change on follow-up digital subtraction angiography (DSA) compared with that of onset. Clipping was performed in 30 cases and coil embolization in 8 cases. IAFC was performed an average of 6.6 days after onset. Color gradient mapping demonstrated reduction of the circulation time after IAFC compared with before IAFC on 39 sides, no change on 15 sides, and extension on 3 sides. Average arterial circulation time before IAFC was 2.25 ± 0.57 s and after IAFC was 1.95 ± 0.55 s. IAFC significantly shortened average arterial circulation ($P = 0.005$). No case developed symptomatic vasospasm after IAFC. IAFC significantly reduced the cerebral circulation time after aneurysmal SAH and might be effective for the prevention of symptomatic vasospasm.

Prophylactic intra-arterial administration of fasudil hydrochloride for vasospasm following subarachnoid haemorrhage.

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We evaluated patients treated with prophylactic intra-arterial administration of fasudil hydrochloride (IAF) after subarachnoid haemorrhage (SAH). Between August 1998 and December 2012, 92 patients with aneurysmal SAH were treated with IAF for angiographic vasospasm without ischemic symptoms after their follow-up angiography. Patients comprised 50 women and 42 men, aged 24-83 (mean 56.6) years. IAF consisted of 15 mg of fasudil hydrochloride dissolved in 20 ml physiological saline and injected through a catheter during approximately 15 min, after diagnostic angiography. The clinical outcome was evaluated using the Glasgow Outcome Scale (GOS) at discharge and ischemic lesions resulting from vasospasm were assessed on computed tomography (CT) scan at discharge. Forty-eight patients underwent surgical clipping and 44 patients underwent endovascular coiling. Angiographic improvement was observed in all patients (100 %). At discharge, 76 (83.0 %) of 92 patients showed good recovery on GOS. Nine patients developed progression of delayed ischemic neurological deficits (DIND) and three of these patients had ischemic lesions on CT scans. No patient had any significant changes in vital signs or any other adverse effects resulting from IAF. IAF therapy was safe and effective for patients with vasospasm following SAH. Prophylactic IAF therapy may prevent symptomatic vasospasm.

Effect of fasudil hydrochloride, a protein kinase inhibitor, on cerebral vasospasm and delayed cerebral ischemic symptoms after aneurysmal subarachnoid hemorrhage.

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The efficacy and safety of fasudil hydrochloride, a novel protein kinase inhibitor, were evaluated for the treatment of cerebral vasospasm and associated cerebral ischemic symptoms in patients with ruptured

cerebral aneurysm. This randomized open trial with nimodipine as the control included 72 patients who underwent subarachnoid hemorrhage surgery for ruptured cerebral aneurysm of Hunt and Hess grades I to IV. For 14 days following surgery, patients were administered either 30 mg of fasudil hydrochloride by intravenous injection over a period of 30 minutes three times a day or 1 mg/hr of nimodipine by continuous intravenous infusion. Fasudil hydrochloride and nimodipine both showed inhibitory effects on cerebral vasospasm. The incidence of symptomatic vasospasm was five of 33 patients in the fasudil group and nine of 32 patients in the nimodipine group. Good recovery evaluated by the Glasgow Outcome Scale was achieved by 23 of 33 patients in the fasudil group and 19 of 34 patients in the nimodipine group. Both drugs significantly improved consciousness levels and neurological deficits such as aphasia. However, fasudil hydrochloride improved motor disturbance more than nimodipine. Adverse reactions occurred in 13 of 37 patients receiving fasudil hydrochloride and 15 of 35 patients receiving nimodipine. There were no serious adverse events in the fasudil group. The results of this clinical trial indicate that fasudil hydrochloride is a safe and efficient agent for suppressing cerebral vasospasm after subarachnoid hemorrhage surgery for ruptured cerebral aneurysm.

Safety and efficacy of fasudil monotherapy and fasudil-ozagrel combination therapy in patients with subarachnoid hemorrhage: sub-analysis of the post-marketing surveillance study.

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Sub-analysis of the fasudil post-marketing surveillance study compared the safety and efficacy of fasudil plus ozagrel to fasudil only. A total of 3690 patients received fasudil and 1138 received fasudil plus ozagrel between 1995 and 2000. The occurrence of adverse events, occurrence of low density areas associated with vasospasm on computed tomography, absence of symptomatic vasospasm, and poor clinical outcomes associated with vasospasm were compared between the fasudil and fasudil plus ozagrel groups. The pharmacokinetics of fasudil were assessed in 5 patients with subarachnoid hemorrhage. The drug interaction between fasudil and ozagrel was pharmacologically investigated in vitro and in vivo. The occurrence of adverse events and clinical outcomes were similar between the two groups. The occurrences of symptomatic vasospasm and low density areas were lower in the fasudil group than in the fasudil plus ozagrel group. The average trough value (8-hour value) of the fasudil active metabolite, hydroxyfasudil, was 50 nM. Fasudil showed no pharmacological interaction with ozagrel. The combination of fasudil plus ozagrel was well tolerated, but did not result in better efficacy than fasudil only.

Intra-arterial infusion of fasudil hydrochloride for treating vasospasm following subarachnoid haemorrhage.

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In this pilot study we treated cerebral vasospasm in patients with subarachnoid haemorrhage to assess intra-arterial fasudil hydrochloride. We analysed effects of intra-arterial infusion on angiographically evident cerebral vasospasm in 10 patients including 3 with symptoms of vasospasm. Over 10 to 30 min 15 to 60 mg was administered via the proximal internal carotid artery or vertebral artery following standard angiography, without superselective techniques. A total of 24 arterial territories (21 internal

carotid, 3 vertebral) were treated. Angiographic improvement of vasospasm was demonstrated in 16 arterial territories (local dilation in 2, diffuse dilation in 14) in 9 patients. In 2 symptomatic patients, intra-arterial fasudil hydrochloride was associated with resolution of symptoms without sequelae. In the third symptomatic patient the benefit of fasudil hydrochloride was only temporary, and a large cerebral infarction occurred. All asymptomatic patients showed no progression of angiographic to symptomatic vasospasm after treatment with intra-arterial fasudil hydrochloride. No adverse effect was encountered.

Fasudil Stimulates Neurite Outgrowth and Promotes Differentiation in C17.2 Neural Stem Cells by Modulating Notch Signalling but not Autophagy.

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Neurite outgrowth is one of the important therapeutic strategies for neuronal plasticity and regeneration in neural disorders. Fasudil is a clinical medication that is used to treat subarachnoid haemorrhage (SAH) and that is beneficial for many animal models of central nervous system (CNS) diseases. In this study, we hypothesised that fasudil administration would promote neurite outgrowth in neural stem cells (NSCs). Changes in cell morphology were imaged under a light microscope, and neurite-bearing cells were counted. Cell viability and the necrosis rate were determined by MTT and LDH assays, respectively. Additionally, western blot and immunofluorescence analyses were performed to detect protein expression levels. We found that fasudil promoted neurite outgrowth in C17.2 cells in a time- and dose-dependent manner. The neurite-bearing C17.2 cells were differentiated by detecting the changes in neural and astrocytic markers after fasudil treatment through down-regulating Notch signalling. Previously, fasudil was reported to induce autophagy, which plays an important role in neural differentiation. However, both rapamycin, an autophagy inducer, and 3-methyl-adenine (3-MA), an autophagy inhibitor, had no effects on the fasudil-induced neurite outgrowth, suggesting that autophagy may be not involved in this process. In summary, fasudil could stimulate neurite outgrowth and differentiation in C17.2 cells by modulating Notch signalling but not autophagy.

Efficacy of fasudil for the treatment of aneurysmal subarachnoid hemorrhage: A systematic review protocol of randomized controlled trial.

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This study aims to systematically assess the efficacy and safety of fasudil for the treatment of aneurysmal subarachnoid hemorrhage (ASH). This study will include all of randomized controlled trials on the efficacy and safety of fasudil for the treatment of ASH. Ten electronic databases of PubMed, Embase, Cochrane Library, Google Scholar, Web of Science, Ovid, Cumulative Index to Nursing and Allied Health Literature, Allied and Complementary Medicine Database, Chinese Biomedical Literature Database, and China National Knowledge Infrastructure will be searched from inception to the May 1, 2019 without language restrictions. We will also search gray literatures to avoid missing any other potential studies. Two authors will independently perform study selection, data extraction and management, and methodologic quality assessment. The primary outcome is limbs function. The secondary outcomes comprise of muscle strength, muscle tone, quality of life, and adverse events. This study will provide a comprehensive literature search on the current evidence of fasudil for the treatment of ASH from primary and secondary outcomes. The results of this study will present evidence to determine whether fasudil is

an effective and safety treatment for patients with ASH. PROSPERO CRD42019136215.

Pleiotropic effects of the rho-kinase inhibitor fasudil after subarachnoid hemorrhage: a review of preclinical and clinical studies.

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There is growing evidence that Rho-kinase contributes to cardiovascular disease, which has made Rho-kinase a target for the treatment of human diseases. To date, the only Rho-kinase inhibitor employed clinically in humans is fasudil, which has been used for the prevention of cerebral vasospasm and subsequent ischemic injury after surgery for subarachnoid hemorrhage (SAH). A number of pathological processes, in particular hemodynamic dysfunctions and inflammatory reactions, are thought to be related in the pathogenesis of delayed cerebral vasospasm and subsequent ischemic injury after SAH. This review focuses on fasudil's pleiotropic therapeutic effects: amelioration of hemodynamic dysfunction and inflammation, and discusses in detail the clinical studies on fasudil administered after the occurrence of SAH.

Efficacy and safety of fasudil in patients with subarachnoid hemorrhage: final results of a randomized trial of fasudil versus nimodipine.

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Fasudil is believed to be at least equally effective as nimodipine for the prevention of cerebral vasospasm and subsequent ischemic injury in patients undergoing surgery for subarachnoid hemorrhage (SAH). We report the final results of a randomized, open trial to compare the efficacy and safety of fasudil with nimodipine. A total of 63 patients undergoing surgery for SAH received fasudil and 66 received nimodipine between 1998 and 2004. Symptomatic vasospasm, low density areas on computed tomography (CT), clinical outcomes, and adverse events were all recorded, and the results were compared between the fasudil and nimodipine groups. Absence of symptomatic vasospasm, occurrence of low density areas associated with vasospasm on CT, and occurrence of adverse events were similar between the two groups. The clinical outcomes were more favorable in the fasudil group than in the nimodipine group ($p = 0.040$). The proportion of patients with good clinical outcome was 74.5% (41/55) in the fasudil group and 61.7% (37/60) in the nimodipine group. There were no serious adverse events reported in the fasudil group. The present results suggest that fasudil is equally or more effective than nimodipine for the prevention of cerebral vasospasm and subsequent ischemic injury in patients undergoing surgery for SAH.

Antivasospastic effects of hydroxyfasudil, a Rho-kinase inhibitor, after subarachnoid hemorrhage.

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We investigated the anti-vasospastic potential of fasudil's active metabolite, hydroxyfasudil, a Rho-kinase

inhibitor, after subarachnoid hemorrhage (SAH) and also its effect on hemorheological abnormalities following cerebral ischemia. Chronic cerebral vasospasm was produced using a two-hemorrhage canine model. On day 7, angiographic vasospasm was observed in all animals, and intravenous administration of hydroxyfasudil ($3 \text{ mg} \cdot \text{kg}^{-1} \cdot 30 \text{ min}^{-1}$) significantly reversed the vasospasm (predose diameter of the basilar artery, $57.9\% \pm 2.0\%$ of the baseline before the injection of blood; postdose diameter, $64.5\% \pm 1.9\%$). The viscosity of whole blood was significantly increased 24 h after 1 h middle cerebral artery occlusion in rats. Hydroxyfasudil (3 and 10 mg/kg, i.p.) significantly decreased blood viscosity. The specificity of hydroxyfasudil was examined against a panel of 17 protein kinases using ELISA analysis. Hydroxyfasudil inhibited Rho-kinase α and β at a concentration of $10 \text{ } \mu\text{M}$ by 97.6% and 97.7%, respectively. No other protein kinase was inhibited with $10 \text{ } \mu\text{M}$ hydroxyfasudil by over 40%. The present results indicate hydroxyfasudil is a selective inhibitor of Rho-kinase. The results also suggest that hydroxyfasudil contributes to the potency of fasudil to prevent cerebral vasospasm and hyperviscosity and suggest the potential utility of hydroxyfasudil as a therapeutic agent for patients with SAH.